Delivery 2

Task 1: Identifying and finding inconsistencies in vision document

- Time spent during inspection: 60 Minutes

Defect #	Location	Defect type	Classification	Description	Status	Date corrected
1	3-3.1, Page 2	Omission	Major	Not mentioned few major stakeholders that were important from system point of view. These stakeholders would be developers, testers, Quebec Traffic department, project sponsors. Developers and testers are the ones who will develop the system but not take part in information gathering.		
2	3-3.2, Page 3	Omission	Major	Admin should also be a user. Not mentioned it as a user in user summary.		
3	3-3.3, Page 3	Overspecification	Minor	Too much detailed specification of the machine solution is mentioned.		
4	3-3.4, Page 4	Noise	Minor	In Scalable, there is no information regarding server usage in the problem world feature.		
5	4-4.2, Page 5	Unfeasibility	Minor	To develop system platform independent takes a lot more effort from developers' point of view and from budget point of view. Project might take more than the scheduled deadline to make it platform independent.		
6	4-4.2, Page 5	Contradiction	Minor	The system might not always support latest operating system as soon as the new OS is out in market.		
7	6-6.2, Page 6	Ambiguity	Minor	Security term can be interpreted in many ways. It can also be taken as security of users, admin and trainers' data and also the accessing the right section by the authenticated users.		

Inconsistency Form:

#	Location	Inconsistency type	Classification	Description	Status	Date corrected
1	2-2.1, Page 1 S1: They must manage schedule if they book **physical coaching** S2: Since they are attending **physical lectures** they cannot see the lecture video, if they have any doubt, as it was a physical lecture.	Terminology Clash	Weak	Here physical coaching and physical lectures are same, but they are given different name in different statements.		
2	2-2.1, Page 2 S3: a web-based platform in which trainer uploads a driving lesson lecture and then **once the video is drafted, it is reviewed by admin for quality purpose** and if everything is perfect then the video will be published, then students can watch this video and can solve various quizzes. 2-2.2, Page 2 provides an online E-learning platform for students, where trainers prepare a video with various quizzes **then admin check the quality of the**	Structure Clash	Weak	Here the main thing is the trainer reviews the video and post it online, this concept has been written in two ways in two different places.		

	video, once he approves it, they post it online and students can view the lectures. Students can watch videos anytime and solve the quizzes and can track their record					
3	3-3.2, Page 3 S5: Once **log-in into the system**, they can watch the video from where they left last time. 3-3.3, Page 3 S6: Anytime user can use the website **just by entering their personal credential**, if they are already registered, otherwise they can register	Structure Clash	Weak	Here, the main structure of the phrase, log-in into the system and getting into the system just by entering their personal credentials. Here both have the same concept structure but are used in two different statements.		
4	3-3.4, Page 4 S7: **Authenticate user**, admin and trainer with separate id and password 4-4.2, Page 5 S8: **Students**	Designation Clash	Weak	Here user and the student are the one thing. Student and user are used in two		

	should be able to login into the system through their unique id and password, same goes for trainer and the admin.			different statements but share the same concept.		

Other comments/recommendations:
No comments or recommendation.

Task 2: Documenting conflicts

Statement	S1	S2	S3	S4	S5	S6	S7	S8	Total
S1	0	0	0	1	0	1	0	0	2
S2	0	0	1	1	1	0	0	0	3
S3	0	1	0	1000	1000	0	0	0	2001
S4	1	1	1000	0	1000	0	0	0	2002
S5	0	1	1000	1000	0	0	0	0	2001
S6	1	0	0	0	0	0	1000	1000	2001
S7	0	0	0	0	0	1000	0	1000	2000
S8	0	0	0	0	0	1000	1000	0	2000
Total	2	3	2001	2002	2001	2001	2000	2000	12010

Task 3: Conflict resolution

The conflicting statements found from task 2 are as follows:

- **S1:** They must manage schedule if they book physical coaching
  S4: provides an online E-learning platform for students, where trainers prepare a video with various quizzes then admin check the quality of the video, once he approves it, they post it online and students can view the lectures. Students can watch videos anytime and solve the quizzes and can track their record

  By using Restore Conflicting Technique, statement 1 can be reframed as, With the introduction of an interactive E-platform, students don’t have to attend physical lectures, they can watch video anytime anywhere.

  By using Drop Lower Priority Statements, statement 1 can be rewritten as, managing time while learning in physical lecture is very important.

- **S1:** They must manage schedule if they book physical coaching
  S6: Anytime user can use the website just by entering their personal credential, if they are already registered, otherwise they can register

  By using Drop Lower Priority Statements, statement 1 can be rewritten as, it is hard to attend physical lecture at any-time and anywhere, as there is some specific time and place of the physical lecture.

  By using Weaken Conflicting Technique, statement 2 can be reiterated as, if the student is of Concordia University, then he/she can access the website anytime just by entering their personal credentials.

- **S2:** Since they are attending physical lectures they cannot see the lecture video, if they have any doubt, as it was a physical lecture.
  S3: a web-based platform in which trainer uploads a driving lesson lecture and then once the video is drafted, it is reviewed by admin for quality purpose and if everything is perfect then the video will be published, then students can watch this video and can solve various quizzes.

  By using Restore Conflicting Technique, statement 1 can be reframed as, With the introduction of an interactive E-platform, students don't have to attend physical lectures, they can watch video anytime anywhere and can rewatch it, if they have any doubts.

  By using Weaken Conflicting Technique, statement 2 can be restated as, a web-based platform in which trainer uploads a driving lesson lecture and then once the video is drafted, it is reviewed by admin for quality purpose and if everything is perfect then the video will be published, then the authenticated Concordia University Students can watch this video and can solve various quizzes.

- **S2:** Since they are attending physical lectures they cannot see the lecture video, if they have any doubt, as it was a physical lecture.
  S4: provides an online E-learning platform for students, where trainers prepare a video with various quizzes then admin check the quality of the video, once he approves it, they post it online and students can view the lectures. Students can watch videos anytime and solve the quizzes and can track their record

  By Specialize conflict source or target, statement 1 can reiterated as, since youth population is attending physical lectures, they cannot see the lecture video again and they cannot solve the doubt, if they have any.

  By using Weaken Conflicting Technique, statement 2 can be restated as, unless students are from Concordia University, they might have to attain physical lectures to learn.

- **S2:** Since they are attending physical lectures they cannot see the lecture video, if they have any doubt, as it was a physical lecture.
  S5: Once log-in into the system, they can watch the video from where they left last time.

  By Weaken Conflicting Statements, statement 2 can rewritten as, By Signing into the system one can access the video where they have left before, unlike the physical lectures where students need to attend physical lectures.

  By specialize conflict source or target, statement 2 can be reframed as, once log-in into the system, Concordia students can watch video online anytime anywhere.

Task 4: Conflict Evaluation

Conflict Evaluation Table for Conflict No. 1:

Alternative Option 1: With the introduction of an interactive E-platform, students don't have to attend physical lectures, they can watch video anytime anywhere.

Alternative Option 2: Managing time while learning in physical lecture is very important.

Evaluation Criteria	Significance Weighting	Alternative Option 1	Alternative Option 2
Time efficient	0.4	0.9	0.3
Minimal Inconvenience	0.2	0.7	0.4
Reliable Response	0.4	0.8	0.4
Total	1.0	0.82	0.36

So, alternate option 1 can be used in place of S1 to resolve the conflict.

Conflict Evaluation Table for Conflict No. 2:

Alternate Option 1: It is hard to attend physical lecture at any-time and anywhere, as there is some specific time and place of the physical lecture.

Alternate Option 2: if the student is of Concordia University, then he/she can access the website anytime just by entering their personal credentials.

Evaluation Criteria	Significance Weighting	Alternative Option 1	Alternative Option 2
Highly Inconvenience	0.3	0.8	0.2
Accessing the System	0.2	0.3	0.8
Time Efficient	0.5	0.2	0.9
Total	1.0	0.4	0.67

So, alternate option 2 can be used in place of S4 to resolve the conflict.

Conflict Evaluation Table for Conflict No. 3:

Alternate Option 1: With the introduction of an interactive E-platform, students don't have to attend physical lectures, they can watch video anytime anywhere and can rewatch it, if they have any doubts.

Alternate Option 2: a web-based platform in which trainer uploads a driving lesson lecture and then once the video is drafted, it is reviewed by admin for quality purpose and if everything is perfect then the video will be published, then the authenticated Concordia University Students can watch this video and can solve various quizzes.

Evaluation Criteria	**Significance Weighting**	**Alternative Option 1**	**Alternative Option 2**
Interactive	**0.4**	**0.7**	**0.7**
Quality	**0.2**	**0.6**	**0.8**
Authentication	**0.3**	**0.6**	**0.9**
Doubt Solving Capacity	**0.1**	**0.8**	**0.6**
Total	**1.0**	**0.66**	**0.77**

So, alternate option 2 can be used in place of S3 to resolve the conflict.

Conflict Evaluation Table for Conflict No. 4:

Alternate Option 1: Since youth population is attending physical lectures, they cannot see the lecture video again and they cannot solve the doubt, if they have any.

Alternate Option 2: Unless students are from Concordia University, they might have to attain physical lectures to learn.

Evaluation Criteria	**Significance Weighting**	**Alternative Option 1**	**Alternative Option 2**
Clarifying who can access the system	**0.7**	**0.4**	**0.8**
Doubt Solving Capacity	**0.3**	**0.2**	**0.2**
Total	**1.0**	**0.34**	**0.62**

So, alternate option 2 can be used to resolve the conflict.

Conflict Evaluation Table for Conflict No. 5:

Alternate Option 1: By Signing into the system one can access the video where they have left before, unlike the physical lectures where students need to attend physical lectures.

Alternate Option 2: Once log-in into the system, Concordia students can watch video online anytime anywhere.

Evaluation Criteria	**Significance Weighting**	**Alternative Option 1**	**Alternative Option 2**
Availability	**0.3**	**0.7**	**0.7**

Authentication	**0.3**	**0.8**	**0.8**
Accessibility	**0.4**	**0.9**	**0.6**
Total	**1.0**	**0.81**	**0.69**

So, alternate option 1 can be used to resolve the conflict.

Task 5: Risk Management

(A) Risk Identification:

Component Inspection:

The possible Risk that can be identified using component Inspection are as follows:

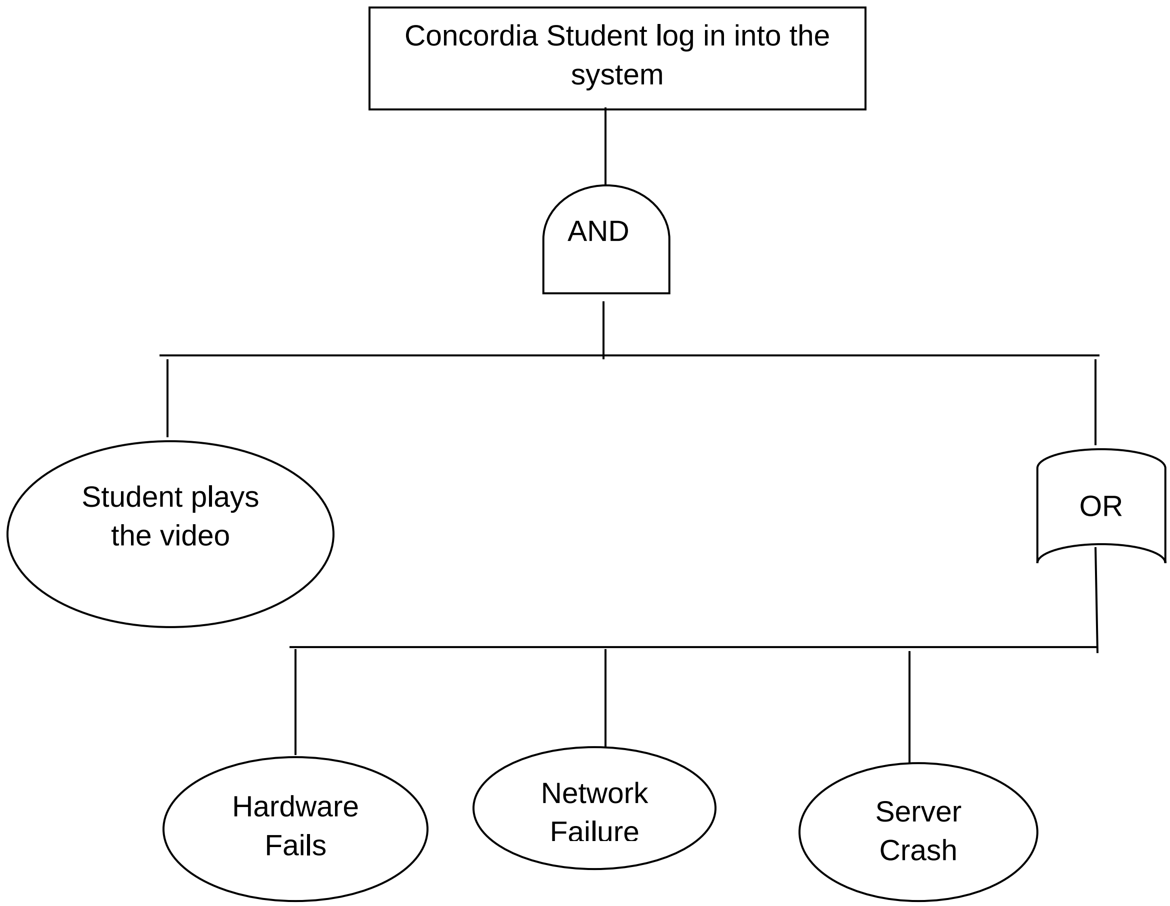
1. **Server System:** Any hacker can hack server and can steal personal information from the databases.
2. **Communication System:** Network of the telecom company can be lost or down anytime under unnatural circumstances.
3. **Hardware System:** Hardware system of users' device from which he/she is accessing the website might get damaged. E.g.: Device Screen, Device RAM, etc.

Risk Checklist:

The possible Risks that can be identified using Risk Checklist are as follows:

1. **Confidentiality and Availability:** Authenticated user's id and password, important and private information of a user can be hacked by various black hat hackers, Cyber Terrorist. Also, hackers keep the authenticate users away from accessing the system by performing DOS attack on the system.
2. **Cost and Deadline:** The overall cost of developing the system might increase by making the system platform independent. Also, deadline of completing the project in time might also increase to make the system platform independent.
3. **Useability:** With so much functionality in the system, user might find it hard to use the system easily.

Risk Tree:



Thus, from this risk tree, and with the help of cut-set, the possible risks identified are: Hardware Failure, Network Failure and Server Crash.

(B) <u>Quantitative Assessment of Risk Identified:</u>

1. Server System / Confidentiality and Availability:

Estimated Cost = $0.3 Million
Probability of this Risk = 0.04

Risk Exposure = impact x Risk Probability
= $0.3m x 0.04
= $12,000

2. **Communication System / Network Failure:**

   Estimated Cost = \$10,000
   Probability of this Risk = 0.01

   Risk Exposure = impact x Risk Probability
   = \$10,000 x 0.01
   = \$100

3. **Hardware System Failure:**

   Estimated Cost = \$8,000
   Probability of this Risk = 0.01

   Risk Exposure = impact x Risk Probability
   = \$8,000 x 0.01
   = \$80

4. **Cost and Deadline:**

   Estimated Cost = \$1 million
   Probability of this Risk = 0.08

   Risk Exposure = impact x Risk Probability
   = \$1m x 0.08
   = \$80,000

5. **Useability:**

   Estimated Cost = \$0.6 million
   Probability of this Risk = 0.03

   Risk Exposure = impact x Risk Probability
   = \$0.6m x 0.03
   = \$18,000

6. **Server Crash:**

   Estimated Cost = \$0.2 million

Probability of this Risk = 0.02

Risk Exposure = impact x Risk Probability
= \$0.2 x 0.02
= \$4,000

(C) <u>Risk Control:</u>

1. **Server System / Confidentiality and Availability:**

Risk Exposure $_{before}$ = \$12,000

Alternative Option 1: By Reduce Consequence likelihood tactic, this risk can be countered by introducing new requirement as, any information, particularly private information, stored into database must be stored in encrypted form.

Estimated Cost = \$0.3 million
Probability of this Risk = 0.03

Risk Exposure = impact x Risk Probability
= \$0.3 x 0.03
= \$9,000

Risk Reduce Leverage (RRL):

Cost of Risk Reduction = \$4000
Risk Reduce Leverage (1) = (RE $_{before}$ – RE $_{after}$)/Cost of Risk Reduction
= (12000 – 9000)/4000
= 0.75

Alternative Option 2: By Reduce Risk likelihood tactic, this risk can be countered by introducing new requirement as, Server should be protected digitally by investing in a firewall, protecting system code, using Secure Sockets Layer (SSL), limiting uploads, and using passwords.

Estimated Cost = \$0.3 million
Probability of this Risk = 0.02

Risk Exposure = impact x Risk Probability
= $0.3 x 0.02
= $6,000

Risk Reduce Leverage (RRL):

Cost of Risk Reduction = $5000
Risk Reduce Leverage (2) = (RE_{before} – RE_{after})/Cost of Risk Reduction
= (12000 – 6000)/5000
= 1.2

Here, while comparing RRL of both the alternatives, option 2 looks more promising because the RRL value of option (2) is greater than 1.

2. **Communication System / Network Failures:**

Risk Exposure $_{before}$ = $100

Alternative Option 1: By Reduce Consequence likelihood tactic, this risk can be countered by introducing new requirement as, Users must have an internet speed of good bandwidth, and should buy service of the best Internet Service Provider in the area.

Estimated Cost = $10,000
Probability of this Risk = 0.01

Risk Exposure = impact x Risk Probability
= $10000 x 0.001
= $10

Risk Reduce Leverage (RRL):

Cost of Risk Reduction = $80
Risk Reduce Leverage (1) = (RE_{before} – RE_{after})/Cost of Risk Reduction
= (100 – 10)/80
= 1.125

Alternative Option 2: By Reduce Risk likelihood tactic, this risk can be countered by introducing new requirement as, Users must use laptops, smart phones which have power storage, so that in power failure too, he/she can be connected to the system.

Estimated Cost = $10,000

Probability of this Risk = 0.01

Risk Exposure = impact x Risk Probability
= \$10,000 x 0.008
= \$80

Risk Reduce Leverage (RRL):

Cost of Risk Reduction = \$100
Risk Reduce Leverage (2) = (RE_{before} – RE_{after})/Cost of Risk Reduction
= (100 – 80)/100
= 0.2

Here, while comparing RRL of both the alternatives, option 1 looks more promising because the RRL value of option (1) is greater than 1.

3. **Hardware System / Hardware Failure:**

Risk Exposure $_{before}$ = \$80

Alternative Option 1: By Avoid Risk tactic, this risk can be countered by introducing new requirement as, Students should upgrade to latest devices to stay away from hardware failure.

Estimated Cost = \$8,000
Probability of this Risk = 0.001

Risk Exposure = impact x Risk Probability
= \$8000 x 0.001
= \$8

Risk Reduce Leverage (RRL):

Cost of Risk Reduction = \$60
Risk Reduce Leverage (1) = (RE_{before} – RE_{after})/Cost of Risk Reduction
= (80 – 8)/60
= 1.2

Alternative Option 2: By Reduce Risk likelihood tactic, this risk can be countered by introducing new requirement as, user should regularly check its hardware maintenance and its compatibility with the system.

Estimated Cost = \$8,000
Probability of this Risk = 0.01

Risk Exposure = impact x Risk Probability
= \$8000 x 0.005
= \$40

Risk Reduce Leverage (RRL):

Cost of Risk Reduction = \$88
Risk Reduce Leverage (2) = (RE $_{before}$ – RE $_{after}$)/Cost of Risk Reduction
= (80 – 40)/88
= 0.455

Here, while comparing RRL of both the alternatives, option 1 looks more promising because the RRL value of option (1) is greater than 1.

4. Cost and Deadline:

Risk Exposure $_{before}$ = \$80,000

Alternative Option 1: By Reduce Risk likelihood tactic, this risk can be countered by introducing new requirement as, initially the system should be developed for any one platform and when it performs well or if demanded, it can be made platform independent.

Estimated Cost = \$1 million
Probability of this Risk = 0.04

Risk Exposure = impact x Risk Probability
= \$1 x 0.04
= \$40,000

Risk Reduce Leverage (RRL):

Cost of Risk Reduction = \$50,000
Risk Reduce Leverage (1) = (RE $_{before}$ – RE $_{after}$)/Cost of Risk Reduction
= (80,000 – 40,000)/50,000
= 0.8

Alternative Option 2: By Avoid Risk tactic, this risk can be countered by introducing new requirement as, System should be developed by experienced developers, who have a vast knowledge of different programming languages, and can work rapidly.

Estimated Cost = \$1 million

Probability of this Risk = 0.02

Risk Exposure = impact x Risk Probability
= $1 x 0.02
= $20,000

Risk Reduce Leverage (RRL):

Cost of Risk Reduction = $40,000
Risk Reduce Leverage (2) = (RE $_{before}$ – RE $_{after}$)/Cost of Risk Reduction
= (80,000 – 20,000)/
= 1.5

Here, while comparing RRL of both the alternatives, option 2 looks more promising because the RRL value of option (2) is greater than 1.

5. **Useability:**

Risk Exposure $_{before}$ = $18,000

Alternative Option 1: By Reduce Risk likelihood tactic, this risk can be countered by introducing new requirement as, system should have a good and simple user interface which should be attractive and easy to use for all kind of people.

Estimated Cost = $0.6 million
Probability of this Risk = 0.02

Risk Exposure = impact x Risk Probability
= $0.6 x 0.02
= $12,000

Risk Reduce Leverage (RRL):

Cost of Risk Reduction = $7,000
Risk Reduce Leverage (1) = (RE $_{before}$ – RE $_{after}$)/Cost of Risk Reduction
= (18,000 –12,000)/
= 0.857

Alternative Option 2: By Avoid Risk tactic, this risk can be countered by introducing new requirement as, System should have a “How-to-use” pdf and a video explaining how to use various features of the system and a good user interface.

Estimated Cost = $0.6 million
Probability of this Risk = 0.01

Risk Exposure = impact x Risk Probability
= $0.6 x 0.01
= $6,000

Risk Reduce Leverage (RRL):

Cost of Risk Reduction = $10,000
Risk Reduce Leverage (2) = (RE_{before} – RE_{after})/Cost of Risk Reduction
= (18,000 – 6,000)/10,000
= 1.2

Here, while comparing RRL of both the alternatives, option 2 looks more promising because the RRL value of option (2) is greater than 1.

6. **Server Crash:**

Risk Exposure $_{before}$ = $4,000

Alternative Option 1: By Risk Consequence likelihood tactic, this risk can be countered by introducing new requirement as, Cloud storage, offsite backups, and site duplication Should be done that can be used to prevent permanent loss of data and operational capabilities.

Estimated Cost = $0.2 million
Probability of this Risk = 0.01

Risk Exposure = impact x Risk Probability
= $0.2 x 0.01
= $2,000

Risk Reduce Leverage (RRL):

Cost of Risk Reduction = $1800
Risk Reduce Leverage (1) = (RE_{before} – RE_{after})/Cost of Risk Reduction
= (4000 – 2000)/1800
= 1.11

Alternative Option 2: By Reduce Risk likelihood tactic, this risk can be countered by introducing new requirement as, good server to be used which can handle a lot more traffic at a single time and regular backup of all data to be done at different servers.

Estimated Cost = $0.2 million
Probability of this Risk = 0.015

Risk Exposure = impact x Risk Probability
= $0.2 x 0.015
= $3,000

Risk Reduce Leverage (RRL):

Cost of Risk Reduction = $1200
Risk Reduce Leverage (2) = (RE_{before} – RE_{after})/Cost of Risk Reduction
= (4000– 3000)/1200
= 0.833

Here, while comparing RRL of both the alternatives, option 1 looks more promising because the RRL value of option (1) is greater than 1.